

BETONIQ WEST LTD

Master Implementation Walkthrough: Agent Zero Self-Hosted Clone + WhatsApp Backup Agent External Hosting Migration

Prepared for: Tom Ugbodaga, Shareholder & CEO

Status: DOCUMENTED, ready to execute once revenue lands

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This is the literal, command-by-command version of the two prior summary documents (Agent Zero Setup Checklist and WhatsApp Backup Agent External Hosting Plan). Follow Part A first, then Part B. Total estimated time once started: 2-4 hours across both. Estimated monthly cost if using the free routes throughout: \$0. Estimated cost on the recommended-for-reliability paid routes: roughly \$10-13/month combined.

PART A — AGENT ZERO SELF-HOSTED CLONE, STEP BY STEP

A1. Decide free vs paid before you start

Free route: your own laptop + Docker + Ollama (local model). \$0 forever, but noticeably less sharp reasoning. Paid route: same laptop/VPS + Docker + an OpenAI or Anthropic API key. Sharper output, billed per use (usually a few dollars/month at light usage). You can start free and switch the model later without rebuilding anything.

A2. Install Docker

1. Go to docker.com/products/docker-desktop and download Docker Desktop for your OS (Windows/Mac).
2. Run the installer, accept defaults, restart your laptop if it asks.
3. Open Docker Desktop once and confirm it says "Docker is running" (green icon).

Verify from a terminal:

```
docker --version  
# should print something like: Docker version 27.x.x
```

A3. Get an OpenAI or Anthropic API key (skip if going 100% free/Ollama)

1. OpenAI route: platform.openai.com/api-keys -> Create new secret key -> copy it somewhere safe.
2. Anthropic route: console.anthropic.com/settings/keys -> Create Key -> copy it.

Skip this whole step entirely if you're doing the free Ollama route below.

A4. (Free route) Install Ollama and pull a model

1. Go to ollama.com/download, install for your OS.
2. Open a terminal and pull a capable free model:

```
ollama pull llama3.1:8b  
# downloads a free local model, several GB, one-time
```

3. Test it runs:

```
ollama run llama3.1:8b  
# type a message, confirm it replies, then type /bye  
to exit
```

A5. Clone and launch Agent Zero

```
git clone https://github.com/agent0ai/agent-zero  
cd agent-zero  
docker compose up -d  
# wait ~1-2 minutes for first pull/build
```

1. Open your browser to <http://localhost:50001> (Agent Zero's default web UI port — check the repo README for the current port if this changed).
2. You should see the Agent Zero chat interface load.

A6. Connect the model

1. In Agent Zero's Settings panel, find the Model/LLM provider section.
2. Free route: select Ollama as provider, point it at <http://localhost:11434> (Ollama's default), choose llama3.1:8b as the model.
3. Paid route: select OpenAI or Anthropic as provider, paste your API key from step A3, choose a model (e.g. gpt-4o or claude-3-5-sonnet).
4. Save settings.

A7. Load the BETONIQ WEST identity into it

1. Open this URL and copy the full contents: https://raw.githubusercontent.com/TOMDEGGS/BETONIQ-WEST-/main/AGENT_

CLONE_777.md

2. In Agent Zero's Settings, find the "Agent Prompt" or "System Prompt / Persona" field.
3. Paste the entire AGENT_CLONE_777.md content into it and save.
4. In the chat box, test with: "What's my title, and what's the FID deadline?" — it should answer "Shareholder & CEO" and reference the September 1, 2026 FID portal reopening.

A8. Give it git-push ability

1. Go to github.com/settings/tokens -> Generate new token (classic) -> tick only the "repo" scope -> Generate -> copy the token immediately (shown once).
2. In Agent Zero's Settings -> Secrets/Environment Variables, add: GITHUB_TOKEN = (the token you just copied).
3. Test in chat: "Clone <https://github.com/TOMDEGGS/BETONIQ-WEST> into your workspace, add a one-line test file, commit, and push it."
4. Check the GitHub repo online to confirm the test commit actually landed.

If step 4 works, Agent Zero is now a genuine hands-on clone: it can read, write, and push code exactly like the primary agent, running entirely on your own machine.

PART B — WHATSAPP BACKUP AGENT EXTERNAL HOSTING, STEP BY STEP

B1. Choose and set up the host

1. Go to render.com (or railway.app as the alternative) and sign up with betoniqwestltd@gmail.com.
2. Both offer a free tier suitable for testing; Render's free web services sleep after 15 minutes idle (adds a few seconds delay on first message after sleep) — fine for testing, worth upgrading to the paid tier (~\$7/month) later for instant replies at all times.

B2. Rebuild the backup agent as a standalone app

Create a new folder locally with these two files, replicating exactly what `whatsappBackupAgent` already does today on Base44:

```
// package.json
{
  "name": "betoniq-whatsapp-backup",
  "version": "1.0.0",
  "main": "server.js",
  "dependencies": {
    "express": "^4.19.2",
    "openai": "^4.55.0",
    "twilio": "^5.2.2"
  }
}
```

```
// server.js (part 1 of 2)
const express = require('express');
const OpenAI = require('openai');
const twilio = require('twilio');
const app = express();
app.use(express.urlencoded({ extended: false }));

const openai = new OpenAI({ apiKey:
process.env.OPENAI_API_KEY });
const memory = {}; // phone -> [ {role, content}, ...
] in-memory for test;
// swap for hosted Postgres/SQLite before going live

const SYSTEM_PROMPT = `<< paste full
AGENT_CLONE_777.md content here >>`;
```

```
// server.js (part 2 of 2)
app.post('/whatsapp', async (req, res) => {
  const from = req.body.From;
  const body = req.body.Body;
  memory[from] = memory[from] || [];
  memory[from].push({ role: 'user', content: body });

  const completion = await
  openai.chat.completions.create({
    model: 'gpt-4o-mini',
    messages: [{ role: 'system', content: SYSTEM_PROMPT },
    ...memory[from]],
  });
  const reply = completion.choices[0].message.content;
  memory[from].push({ role: 'assistant', content: reply
  });

  const twiml = new twilio.twiml.MessagingResponse();
  twiml.message(reply);
  res.type('text/xml').send(twiml.toString());
});

app.listen(process.env.PORT || 3000);
```

This mirrors the exact logic already live and tested in the Base44 whatsappBackupAgent function — same system prompt approach, same Twilio TwiML reply pattern.

B3. Push this small app to its own GitHub repo

```
git init
git add .
git commit -m "Initial standalone WhatsApp backup
agent"
git remote add origin https://github.com/TOMDEGGS/beto
niq-whatsapp-standalone
git push -u origin main
```

B4. Deploy to Render

1. In Render dashboard: New -> Web Service -> connect the GitHub repo from B3.
2. Build command: `npm install` Start command: `node server.js`
3. Under Environment, add `OPENAI_API_KEY` = (your key).
4. Deploy. Once live, Render gives you a URL like `https://betoniq-whatsapp-standalone.onrender.com`

B5. Add persistent memory (before going live for real)

1. In Render: New -> PostgreSQL (free tier available).
2. Swap the in-memory ``memory`` object in `server.js` for simple `INSERT/SELECT` queries against a `messages` table (phone, role, content, `created_at`).
3. Redeploy.

B6. Test with a temporary number before touching the real one

1. In Twilio Console -> Messaging -> Try WhatsApp -> use the Twilio Sandbox number first.
2. Set that sandbox's webhook to `https://betoniq-whatsapp-standalone.onrender.com/whatsapp`
3. Join the sandbox from your phone (follow Twilio's on-screen join code instructions) and send a test message.
4. Confirm it replies correctly and remembers context across 2-3 messages.

B7. Go live on the real number

1. Twilio Console -> your actual WhatsApp-enabled number -> Messaging Configuration.
2. Change "A message comes in" webhook from the current base44.app URL to the new Render URL from B4.
3. Send yourself a real test message and confirm the reply.
4. Keep the old Base44 whatsappBackupAgent function untouched for 1-2 weeks as a fallback, then retire it once you're confident the new one is stable.

Result: this number now survives a full Base44 outage — it depends only on Render/Railway and OpenAI, both fully independent of Base44.

Final Verification Checklist

- Agent Zero answers company questions correctly using AGENT_CLONE_777.md
- Agent Zero can push a real commit to GitHub on request
- Standalone WhatsApp app deployed and reachable at its Render/Railway URL
- Test message via Twilio sandbox replies correctly with memory intact
- Real WhatsApp number's webhook re-pointed to the new external URL
- Old Base44 function kept as fallback for a transition period, then retired